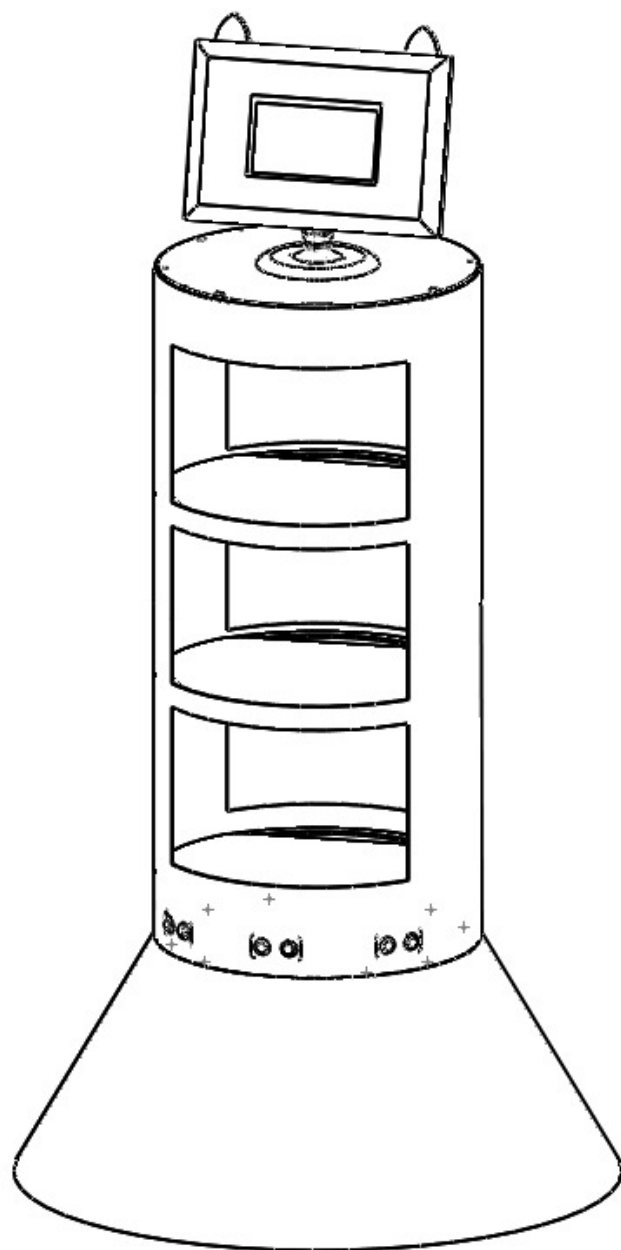


Restaurant Service Robot

Production Manual



Contents

1	Introduction	3
1.1	Purpose of the Document	3
1.2	Target Audience	3
1.3	Project Summary	3
2	Bill of Materials (BOM)	3
3	Mechanical Assembly Instructions	4
3.1	Exploded View Diagrams	4
3.2	Chassis and Frame Assembly	4
3.3	Tray and Mounting Installation	5
3.4	Enclosure Assembly	5
3.5	Assembly Notes	5
4	Electronic Assembly Instructions	5
4.1	PCB Population Guide	6
4.2	Wiring Diagrams and Connectors	6
4.3	Sensor and Actuator Integration	7
4.4	Display and Microcontroller Mounting	7
4.5	Safety Checks Before Power-Up	7
5	PCB Design Details	7
5.1	Board Layout	7
5.2	Gerber Files and Manufacturing Notes	8
6	Firmware and Upload Procedure	8
6.1	Microcontroller Setup	8
6.2	Flashing Instructions	8
6.3	Configuration Parameters	9
6.4	Assembly Drawing with Labels	9
7	Calibration and Configuration	9
7.1	Sensor Calibration	9
8	Quality Control and Checklists	9
8.1	Visual Inspection Checklist	9
8.2	Continuity Test Procedure	9
8.3	Functional Test Checklist	10

9 Packaging and Shipping	10
9.1 Safe Packaging Guidelines	10
10 Appendices	10
10.1 3D Models and Drawings	10
10.2 Source Code Repository Link	11
10.3 Data Sheets and Relevant Documents	11

1 Introduction

1.1 Purpose of the Document

This document serves as the comprehensive production guide for the Restaurant Robot developed under the EN2160 – Electronic Design Realization course. It provides all the technical information necessary for reproducing, assembling, wiring, programming, and testing the robot. The documentation is intended to support manufacturing, maintenance, and future development activities.

1.2 Target Audience

This document is intended for:

- Engineers and technicians involved in hardware assembly and PCB manufacturing
- Personnel responsible for mechanical fabrication and system integration
- Developers maintaining or modifying the firmware and software systems
- Future project teams or manufacturers looking to replicate or upgrade the robot

1.3 Project Summary

The Restaurant Robot is an autonomous indoor service platform designed to transport food and interact with customers in restaurant environments. Inspired by commercial service robots like BellaBot, the system integrates mechanical mobility, sensor-driven navigation, user interaction via a touch display, and modular subsystems communicating through RS-232.

The robot consists of a fabricated steel and plastic chassis with multiple food trays, a SAM3X8E microcontroller at its core, custom PCB for electronics integration, and dedicated firmware for navigation and interaction.

2 Bill of Materials (BOM)

The table below summarizes the key electronic components used in the PCB design of the Restaurant Robot.

Component	Designators	Description	Qty
100nF Capacitors	C1–C2, C8–C34	MLCC 0805, X7R 16V	25
12pF Capacitors	C3–C6	MLCC 0805, C0G	4
2.2uF Capacitors	C7, C12	MLCC 0805, X5R	2
4.7uF Capacitors	C15, C37	MLCC 1206, X7R	2
10uF Capacitors	C24, C35, C41	MLCC 1206, X5R	3
22uF Capacitors	C38, C39	MLCC 1206, X5R	2
Resistors 10k	R1–R15	Chip Resistors 0805	15
Microcontroller	U1	ATSAM3X8E LQFP-144	1
Voltage Regulators	U2, U9	LDL1117 3.3V, SOT-223	2
Flash Memory	U3	W25Q128FVSIQ, 128Mbit SPI	1
Level Shifters	U4–U7	SN74LVC2T45, 2-bit dual	4
Buck Converter	U8	AP63205WU-7, 5V 2A	1
Crystals	Y1, Y2	12MHz, 32.768kHz	2
USB Connector	J1	Micro-USB B, SMD	1
Connectors	J2–J13	JST/XH 2–8 pin	12
Inductor	L1	IHLP2525, 4.7uH	1
LEDs	DS1, DS2	Green SMD LED	2
Test Points	TP1–TP5	Red PhosBronze THM	5
Push Buttons	SW1, SW2	Tactile SPST-NO	2
ESD Protection	RV1	TVS Diode / ESD Chip	1
DC Jack	P1	Barrel Jack Connector	1

Table 1: Simplified Bill of Materials for Restaurant Robot PCB

3 Mechanical Assembly Instructions

This section outlines the step-by-step procedure for assembling the mechanical components of the Restaurant Robot. It includes subassemblies, required tools, and special precautions to ensure mechanical integrity and modularity.

3.1 Exploded View Diagrams

Refer to the attached exploded view drawings from the SolidWorks model for a visual guide of the mechanical structure. Each major part is labeled and cross-referenced with the Bill of Materials (BOM).

3.2 Chassis and Frame Assembly

1. Begin with the steel base frame.
2. Fix the motor mounting brackets onto the rear side using L-angle supports. Align the holes with pre-drilled motor baseplates.
3. Mount the two drive wheels on either side, connected to the NEMA 23 motors.

4. Use vibration-dampening rubber pads under each motor mount to reduce mechanical noise.

3.3 Tray and Mounting Installation

1. Slide the metal trays into their designated rails. Each tray should support up to 5 kg.
2. Ensure adequate spacing between trays for dish clearance.

3.4 Enclosure Assembly

1. Attach the plastic display cover on the top of the robot.
2. Install the 7.0" TFT LCD panel inside the enclosure with proper alignment to the front window.
3. Add wire management clips inside the enclosure to route power and signal lines cleanly.
4. Close the top housing with the plastic enclosure cover. Fix it with screws or clips, ensuring accessibility for maintenance.

3.5 Assembly Notes

- Double-check alignment before tightening screws to prevent misalignment of rotating parts.
- Use thread-lock where necessary to reduce loosening due to vibration.
- Avoid overtightening plastic parts to prevent cracking.
- Conduct a final shake-test to identify any loose mechanical parts.

4 Electronic Assembly Instructions

This section provides step-by-step instructions for assembling the electronic components of the robot. It includes guidance for installing the custom PCB, wiring external modules, and ensuring proper power and signal connections.

4.1 PCB Population Guide

1. Place the PCB on a static-free work surface.
2. Begin with the smallest and lowest-profile components first (e.g., resistors, capacitors).
3. Solder the ICs, connectors, and regulators as per the silkscreen labels and the PCB assembly diagram.
4. Install the RS-232 transceiver, voltage regulators, and power filtering components carefully to avoid overheating.
5. Visually inspect all solder joints and verify for any cold joints or solder bridges.

4.2 Wiring Diagrams and Connectors

- Use labeled jumper wires or ribbon cables to connect external modules to the appropriate headers on the PCB.
- Secure connectors with locking clips or heat-shrink tubing where applicable.
- Follow the wiring as shown in the reference image.

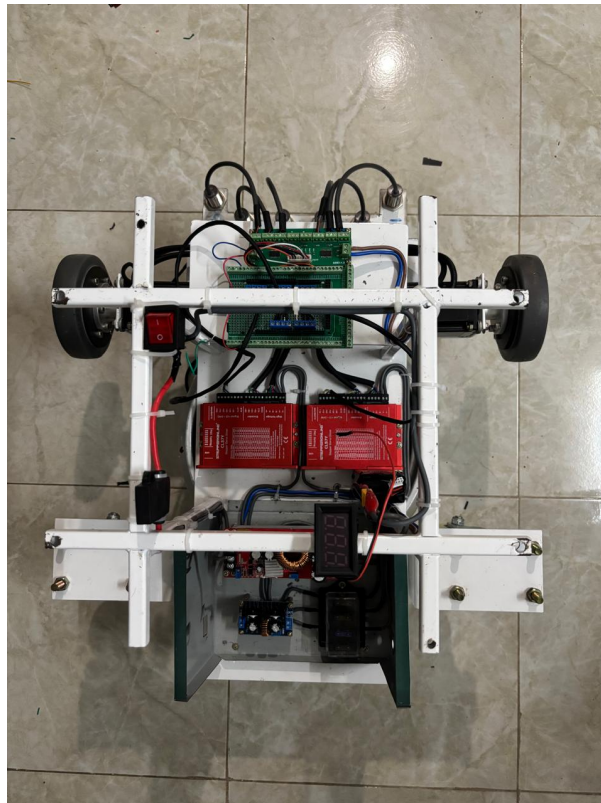


Figure 1: Reference wiring diagram of the PCB with peripherals

4.3 Sensor and Actuator Integration

1. Mount the metal detection sensors and obstacle detection modules at their designated locations on the frame.
2. Run wires back to the PCB using cable sleeves or conduits to avoid tangling with moving parts.
3. Connect sensor outputs to labeled analog/digital input headers on the PCB.
4. Connect the NEMA 23 stepper motors to the motor driver output terminals.

4.4 Display and Microcontroller Mounting

1. Mount the 7.0" TFT LCD to the front panel using standoffs and M4 screws.
2. Connect the 16-bit parallel interface ribbon cable between the display and the microcontroller.
3. Install the SAM3X8E board onto the designated standoffs inside the enclosure.
4. Double-check the microcontroller power, ground, and programming interface connections.

4.5 Safety Checks Before Power-Up

- Verify all ground connections are secure.
- Confirm no shorts between power rails and ground.
- Ensure all high-current paths use proper gauge wires.
- Inspect for correct orientation of all polarized components (diodes, capacitors, ICs).

5 PCB Design Details

5.1 Board Layout

There are 2 PCBs for this robot. Main PCB and the PCB for Proximity Sensors. The Main PCB is designed to be compact and modular, housing the SAM3X8E microcontroller, motor driver interface, power regulators, RS-232 transceiver, and connectors for display, sensors, and audio. Each functional section is isolated for signal clarity.

The main PCB is a 4-layer board.

This stackup enhances EMI performance, reduces ground bounce, and allows better thermal dissipation for power components.

5.2 Gerber Files and Manufacturing Notes

Gerber files were exported for a 4-layer stack with impedance control enabled. The drill file, paste mask, and silkscreen layers are included. Manufacturing parameters:

- Minimum trace width/spacing: 6 mil / 6 mil
- Minimum via size: 0.3 mm
- Board dimensions: approx. 100 mm × 80 mm
- Soldermask: Green, Top and Bottom
- Silkscreen: Top layer only

6 Firmware and Upload Procedure

6.1 Microcontroller Setup

The main controller used in this project is the **ATSAM3X8E** ARM Cortex-M3 microcontroller. It handles sensor data acquisition, display management, RS-232 communication, and actuator control. The microcontroller is mounted on a custom PCB with access to UART, SPI, I2C, and JTAG interfaces. Power is supplied via an onboard 3.3V regulator, and reset/erase pins are routed to pushbuttons for programming convenience.

6.2 Flashing Instructions

Firmware was compiled using a standard ARM-GCC toolchain and flashed to the device using an RS-232 interface. In most cases, the following general steps were followed:

1. Connect the microcontroller to the PC using a standard RS-232 serial cable.
2. Press and hold the ERASE and NRST buttons in sequence to enter bootloader or programming mode.
3. Use a flashing utility (e.g., BOSSA, Atmel Studio, or SAM-BA) to upload the compiled ‘.bin’ or ‘.hex’ file.
4. Once uploaded, perform a reset to run the firmware.

Details may vary based on the specific programmer used and development environment.

6.3 Configuration Parameters

Most configuration parameters such as RS-232 baud rate, sensor thresholds, and display mode settings are defined within the firmware source code. These values can be modified in the configuration header files or adjusted at compile-time.

6.4 Assembly Drawing with Labels

All components are labeled with designators on the silkscreen: resistors (R1–R15), capacitors (C1–C42), connectors (J1–J13), ICs (U1–U9), and jumpers. Component placement ensures easy soldering and access to debugging headers (e.g., JTAG, ERASE, NRST).

7 Calibration and Configuration

7.1 Sensor Calibration

The metal detection sensors and ultrasonic distance sensors are tested using known metallic paths and fixed obstacles. Calibration includes:

- Verifying detection range consistency
- Checking for electromagnetic interference on signal lines

8 Quality Control and Checklists

8.1 Visual Inspection Checklist

- All components correctly placed and soldered
- No visible solder bridges
- Polarized components (diodes, caps, ICs) in correct orientation
- PCB not warped or scratched

8.2 Continuity Test Procedure

1. Set multimeter to continuity mode
2. Probe VCC and GND traces for shorts
3. Verify connection from SAM3X8E pins to connectors (e.g., TX/RX to RS-485, DB pins to display)
4. Confirm power supply paths: 12V input, 5V/3.3V regulators

8.3 Functional Test Checklist

- Display powers on and shows interface
- Metal sensor detects aluminum strip
- Ultrasonic sensor detects hand within 30 cm
- Motors respond to direction commands
- RS-232 transmits and receives messages correctly

9 Packaging and Shipping

9.1 Safe Packaging Guidelines

To protect the robot during transport:

- Place robot in ESD-safe foam-lined box
- Secure moving parts (wheels, trays) with zip ties or tape
- Cover display screen with plastic film
- Label the box with “Fragile” and “This Side Up”

10 Appendices

10.1 3D Models and Drawings

Figure 2 shows the exploded view of the assembled robot, illustrating the mechanical structure, component placement, and enclosure design based on the SolidWorks model.

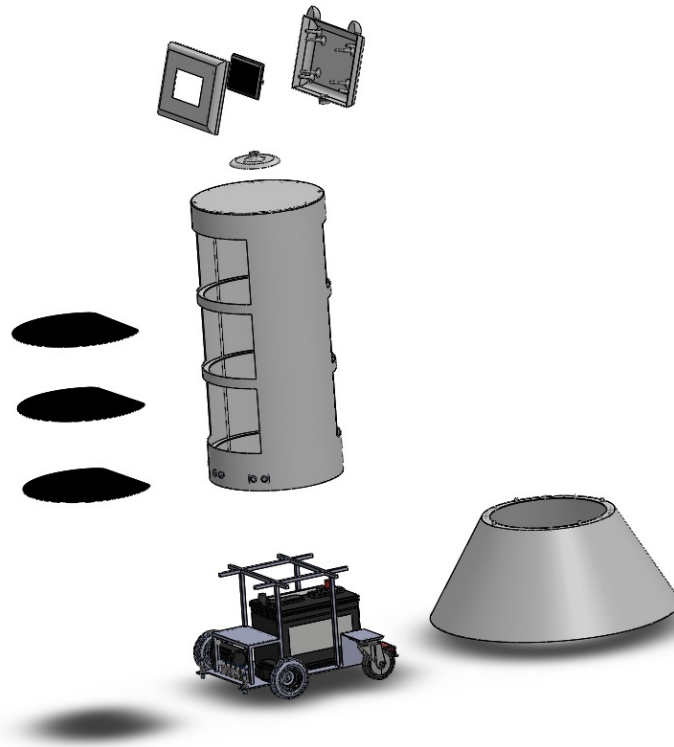


Figure 2: Exploded view of the restaurant robot assembly

10.2 Source Code Repository Link

The complete source code, including firmware, hardware schematics and PCB layout files for the Restaurant Robot project, is hosted on GitHub and can be accessed using the following link:

- <https://github.com/YourUsername/restaurant-robot>

10.3 Data Sheets and Relevant Documents

- Nema 23 Motor and Driver: https://drive.google.com/drive/folders/1t_PN02JcCgiCsATQyiluq-kj0ZTv335-?usp=drive_link
- AO2 PWM Ultrasonic Sensors: https://drive.google.com/drive/folders/1_Xdv5Zs0F8AB279o9QYAdrCsSYbcH0dw?usp=drive_link
- Proximity Sensors: https://drive.google.com/drive/folders/1keqV7BNKRRuUt0pEe417E9VMuuxMAJQu?usp=drive_link
- 7-inch TFT LCD Display: https://drive.google.com/drive/folders/14D8buxf1hrKpHv2f4uz1sQXVDi9iKZ4L?usp=drive_link

- **SAM3X8E** MCU: https://drive.google.com/file/d/1gG4gJJS6T9XMbzaYF70B0jIM091cdNFk/view?usp=drive_link
- **Wheel:** https://drive.google.com/drive/folders/1SpT7xqD51tSb7tsyiDhPFohJYaKiCV0P?usp=drive_link